

Table 1: Link prediction results on Sports-Copurchase and Cloth-Copurchase. All entries are test metrics reported as percentages. Sports-Copurchase is the formal `unified_sampled_lp_v1` benchmark with seeds 42, 43, and 44; values are mean $\pm$ population standard deviation and checkpoint selection uses validation MRR. Cloth-Copurchase values are included for completeness, but are not a second formal LP benchmark. Validation MRR is omitted.

Model	Sports-Copurchase				Cloth-Copurchase			
	MRR	H@1	H@3	H@10	MRR	H@1	H@3	H@10
MLP	24.99 $\pm$ 0.35	11.76 $\pm$ 0.31	26.32 $\pm$ 0.34	55.39 $\pm$ 0.70	18.97 $\pm$ 0.20	9.02 $\pm$ 0.15	19.12 $\pm$ 0.21	39.41 $\pm$ 0.46
GCN	32.09 $\pm$ 0.14	17.77 $\pm$ 0.18	35.44 $\pm$ 0.05	64.11 $\pm$ 0.23	20.99 $\pm$ 0.12	11.04 $\pm$ 0.10	21.92 $\pm$ 0.14	41.05 $\pm$ 0.70
GraphSAGE	32.59 $\pm$ 0.24	17.10 $\pm$ 0.16	36.41 $\pm$ 0.26	69.05 $\pm$ 0.41	24.68 $\pm$ 0.13	12.71 $\pm$ 0.10	26.15 $\pm$ 0.12	50.40 $\pm$ 0.22
MMGCN	34.13 $\pm$ 0.30	18.02 $\pm$ 0.33	39.19 $\pm$ 0.26	71.95 $\pm$ 0.30	25.65 $\pm$ 0.04	13.22 $\pm$ 0.03	27.81 $\pm$ 0.10	53.32 $\pm$ 0.17
MGAT	32.61 $\pm$ 0.11	17.17 $\pm$ 0.05	36.29 $\pm$ 0.26	69.11 $\pm$ 0.61	25.22 $\pm$ 0.13	13.16 $\pm$ 0.08	26.93 $\pm$ 0.23	51.26 $\pm$ 0.30
DGF	34.19 $\pm$ 0.31	18.57 $\pm$ 0.25	38.51 $\pm$ 0.45	70.44 $\pm$ 0.44	<u>25.93 $\pm$ 0.13</u>	<u>13.79 $\pm$ 0.14</u>	27.80 $\pm$ 0.14	51.96 $\pm$ 0.24
DMGC	32.93 $\pm$ 1.77	17.54 $\pm$ 1.57	37.33 $\pm$ 2.43	68.05 $\pm$ 2.33	23.35 $\pm$ 0.11	11.52 $\pm$ 0.07	24.70 $\pm$ 0.08	48.63 $\pm$ 0.24
LGMRec	32.92 $\pm$ 0.03	18.16 $\pm$ 0.08	36.71 $\pm$ 0.14	66.25 $\pm$ 0.09	22.16 $\pm$ 0.20	11.47 $\pm$ 0.22	23.33 $\pm$ 0.28	44.28 $\pm$ 0.29
DiP	<u>35.14 $\pm$ 0.21</u>	<u>18.90 $\pm$ 0.39</u>	<u>40.29 $\pm$ 0.36</u>	<u>72.28 $\pm$ 0.52</u>	25.76 $\pm$ 0.82	13.29 $\pm$ 0.62	<u>27.84 $\pm$ 1.03</u>	52.78 $\pm$ 1.35
CoSI-MAG	37.48 $\pm$ 0.34	21.56 $\pm$ 0.39	43.01 $\pm$ 0.44	73.04 $\pm$ 0.38	27.91 $\pm$ 0.35	15.43 $\pm$ 0.19	30.49 $\pm$ 0.45	54.57 $\pm$ 0.77